

Algorithmic Objectivity for Congressional Redistricting: A National-Scale Demonstration

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Abstract

Congressional redistricting in the United States suffers a legitimacy crisis: partisan mapmakers manipulate district boundaries to predetermine election outcomes, eroding public confidence in representative democracy. We evaluate recursive graph bisection—a computational method used for supercomputer workload distribution—as a partisan-input-excluded geographic benchmark. The execution stage does not consume partisan data, but structure, weights, resolution, seed, and legal modifications remain public policy choices. Governed NRS v0.3 runs assign all 27,398,654 Census blocks across the 2000, 2010, and 2020 cycles. Each cycle covers 50 States, 435 districts, and 385 recursive nodes with independently verified assignment, connectivity, and population-tolerance conformance. Exact weighted-boundary and canonical proof coverage remains zero of 1,155 nodes. Historical demographic, compactness, and partisan analyses belong to earlier tract-based profiles and are not outcome claims for NRS v0.3. A separately governed 2020 common-block bakeoff now reports assignment overlap, subdivision splits, and four block-projected geometry metrics for NRS v0.3 and official CD118 assignments; it does not establish original-linework compactness or plan superiority. A newly implemented public ceremony commits the rule before the root cut and releases certified cuts by tree depth with fixed review windows and independently verifiable prefixes; externally witnessed bakeoff evidence for that process remains pending. The contribution is therefore a reproducible operational baseline and research program, not proof of a uniquely fair map or a substitute for commissions, courts, civil-rights analysis, or public judgment.

1 The Redistricting Crisis

Every ten years, American democracy faces a constitutional requirement with no constitutional solution: redrawing 435 congressional district boundaries to reflect population changes from the decennial census. This process, affecting 330 million Americans and determining electoral outcomes for a decade, suffers from a fundamental legitimacy crisis. Partisan legislatures manipulate district boundaries to predetermine election results—a practice known as gerrymandering—eroding public confidence in representative government [1].

The problem is ancient but newly precise. While gerrymandering dates to 1812, modern computing power enables manipulation with surgical accuracy [2]. Reformers have proposed independent redistricting commissions to remove partisan incentives, yet these still rely on human judgment about what constitutes “fair” districts. Even well-intentioned commissioners must balance competing values—compactness, community preservation, competitiveness—with no objective formula to guide their choices [3].

The legal landscape offers no remedy. In *Rucho v. Common Cause* (2019), the Supreme Court ruled that partisan gerrymandering claims present “political questions beyond the reach of federal courts,” effectively removing federal oversight [4]. Chief Justice Roberts acknowledged the practice “incompatible with

democratic principles” but declared courts institutionally incapable of policing it. This ruling created a governance vacuum: states may gerrymander with impunity, and no mathematical standard exists to distinguish legitimate from illegitimate maps.

This program is scoped to congressional and state-legislative redistricting. Local redistricting for city councils, school boards, and county commissions is outside the present research boundary.

We evaluate recursive graph bisection—a computational method developed for distributing workloads across supercomputer processors—as a reproducible geographic benchmark for redistricting. Applied to all 50 states across three census decades (2000, 2010, 2020), the historical pipeline produces population-balanced, contiguous plans while excluding partisan data from the benchmark execution inputs. This is an input-exclusion claim, not an impossibility theorem: structure, geographic resolution, weights, seed rules, and legally required modifications can affect political outcomes and can themselves be selected strategically. Compact benchmark maps can also reflect geographic voter concentration, as shown in Track C.5. The contribution is an auditable comparison procedure that can support commissions, courts, and public review rather than replace them.

2 The Huntington-Hill Precedent

Congressional redistricting is not the first time American democracy has confronted the need for algorithmic objectivity. For 150 years, Congress struggled with an analogous problem: apportioning House seats among states based on decennial census counts. From 1790 to 1941, every census triggered political battles over which mathematical method to use, with each formula favoring different states. Congress changed the apportionment method six times, always to advantage particular political coalitions [5].

In 1941, Congress adopted the Huntington-Hill method of equal proportions, a formula that has supplied a durable operational allocation rule for eight decades [6]. The method did not end disputes over House size, census counts, constitutional interpretation, or alternative apportionment axioms. Its narrower achievement was to remove cycle-specific allocation discretion after Congress selected the method and inputs.

This precedent suggests a pathway for redistricting. Just as apportionment answered “How many seats per state?” with mathematical objectivity, redistricting might answer “Where to draw district boundaries?” through algorithmic methods. The parallel is instructive:

Apportionment (operational rule)	Redistricting (open rule)
How many seats per state?	Where to draw boundaries?
Mathematical formula (1941)	Partisan negotiation
Durable post-enactment procedure	Persistent method and boundary conflict

The relevant question is whether Congress can create a similarly durable post-enactment decision procedure for a harder problem while preserving legal and representational judgment.

3 Recursive Bisection Method

Recursive graph bisection, originally developed for distributing computational workloads across parallel processors [7], can produce a reproducible geographic benchmark for redistricting. The method represents a state as a graph, balances population while partitioning the graph, and records the resulting assignments and parameters for independent review.

3.1 Benchmark Execution

The historical national pipeline used census tracts as graph nodes and tract population as the vertex weight. Edges connect units that share a published geographic boundary. In the default geographic profile, the edge weight is the shared-boundary length from the applicable TIGER/Line vintage. METIS minimizes the weighted cut while recursively allocating the required number of districts.

This profile excludes election results, party registration, incumbency, candidate identity, race, and community submissions from benchmark generation. That exclusion is auditable from the configuration and manifest, but it does not make the profile value-free. Tract resolution, recursive structure, boundary weighting, population tolerance, random seed, and engine version can all affect the output.

The governed operational profile is NRS v0.3. It uses 15-digit Census blocks, versioned State adjacency contexts, a fixed floor/ceiling recursive seat schedule, manifest-derived seeds, deterministic candidate refinement, and one-based canonical district labels. For each Census cycle, the published package binds the executable, input inventory, standard and legal profiles, State assignment/tree manifests, national proof-coverage summary, and an independent verification pass. NRS v0.3 is a later research profile; it does not silently amend the v0.1 schedule incorporated by the candidate model statute.

Race-conscious VRA-aligned weighting and partisan-weighted analysis are separate research or case-specific profiles. They are not part of the default geographic benchmark and must not be used to describe that benchmark as both race-blind and VRA-compliant.

3.2 Reproducibility And Configuration

A reproducible run must identify:

- census and TIGER/Line releases;
- geographic-unit index and adjacency definition;
- district count and population source;
- structure, edge-weight, search, and engine profiles;
- balance settings and seed;
- source commit, compiler, dependency lock, and binary provenance; and
- canonical hashes linking assignments, analysis, and reports.

The current project supports several structures, weight signals, and search modes. Results from one profile do not establish properties of another. The candidate national-standard decision record selects standard recursive bisection, geographic boundary weights, and one precommitted content-derived seed as a comparison benchmark. Seed wiring and block-level operational conformance are implemented and verified nationally in NRS v0.3. The earlier tract outputs analyzed elsewhere in this paper remain separate research artifacts; they are not NRS v0.3 certificates and cannot supply missing demographic, partisan, or community evidence for the block-level assignments. Geometric compactness is supplied only by the later precommitted 2020 bakeoff, which dissolves NRS and official CD118 assignments over the same retained Census-block polygons. It measures block-projected geometry rather than the official plan’s original polygon linework.

3.3 Sequential Public-Cut Ceremony

For a certified recursive plan, precommitment can be made observable rather than merely asserted after completion. A genesis artifact binds the rule profile, root instance, canonical floor/ceiling district tree, review interval, witness threshold, and challenge policy. The root cut is published after one review window; subsequent non-leaf cuts are released by tree depth and ordered by binary path. Each release includes its split instance, certificate, proof, timestamp, witness receipts, and prior-round hash.

The public verifier checks each disclosed prefix without access to unreleased descendants, including exact parent-to-child universe derivation. Completion binds the final certified tree; a sustained challenge instead leaves an immutable halted transcript. The implementation checks receipt presence and uniqueness, not an external service’s signature or clock, so production claims require separately validated transparency-log or trusted-timestamp adapters.

Prospective bakeoffs use a separate procedural scoreboard for proof coverage, prefix verification, ceremony status, transcript size, verifier time, and negative-test resistance. These fields are not blended with map-quality metrics into a single winner. Existing bakeoffs remain governed by their original frozen protocols and carry no retroactive ceremony claim.

3.4 Population And Contiguity

Population equality is reported as an observed property of each output, not as a claim that one configured tolerance is always legally sufficient. For congressional districts, the controlling legal principle is equality as nearly exact as practicable. A complete evidence package must report ideal population, maximum and total deviation, and any post-partition repair.

Contiguity is evaluated against the published adjacency graph. Water, islands, point contacts, slivers, and bridge corrections can alter that graph and must be versioned rather than silently repaired.

3.5 Performance And Evidence Scope

The Rust implementation makes national experimentation practical, but runtime depends on state size, engine, data cache, profile, and search mode. Historical benchmark tables in this paper are engineering measurements tied to their named environment; they are not universal runtime guarantees.

Ensemble generation is distinct from producing one benchmark plan. A single run can be reproduced and compared, but it cannot define the distribution of valid alternatives. Ensemble percentile claims require archived real traces, multiple chains, convergence diagnostics, immutable election inputs, and uncertainty reporting.

3.6 Input-Exclusion Defense

The defensible procedural claim is narrow: a run whose manifest binds only population and geographic inputs cannot directly target partisan outcomes during execution using partisan data that it never receives. This is an input-exclusion defense, not an impossibility defense.

Partisan consequences may still arise from political geography, and upstream choices can be manipulated if they are selected after observing outcomes. A credible standard therefore precommits assignment-affecting profiles, publishes the benchmark, reports sensitivity, and discloses every baseline-to-final modification. Human legal and community judgment remains necessary.

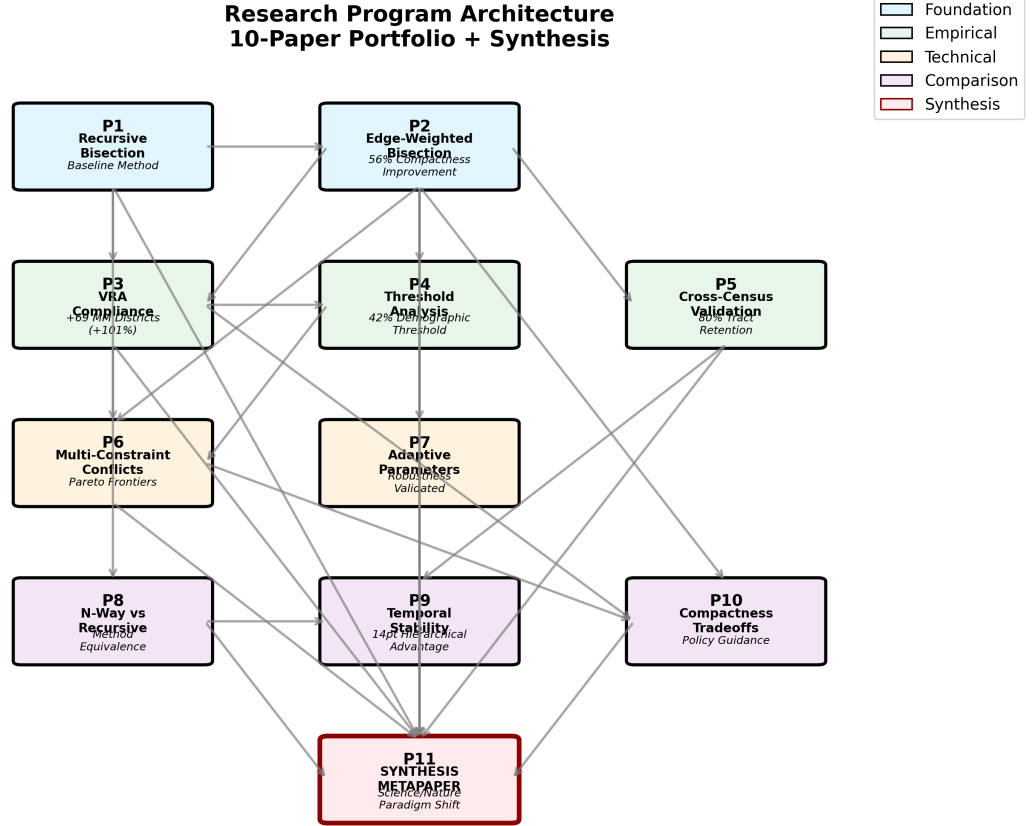


Figure 1: **Research program architecture.** The current index contains 206 papers across 23 active tracks. Foundation papers (blue) establish methods, analysis papers (green) test findings across multiple dimensions, and theory papers (orange) explore algorithmic variations. Internal review and PDF availability do not imply external peer review or complete empirical evidence.

4 National-Scale Findings

Governed NRS v0.3 recursive-bisection runs cover all 50 States across three Census decades (2000, 2010, 2020), generating 1,305 district assignments from 27,398,654 block assignments. The operational counts, structural comparison, and proof gaps below are hash-bound and independently verified. The governed 2020 bakeoff adds same-vintage assignment, subdivision, and block-projected geometry comparisons with official CD118. Regional, demographic, partisan, and older compactness values later in this section remain tied to earlier tract-based research artifacts and must not be read as NRS v0.3 outcomes.

4.1 Regional Context

Algorithmic redistricting performance varies substantially by U.S. Census region, reflecting differences in demographic composition, geographic clustering, and enacted plan characteristics. Table 1 presents key metrics by region (Northeast, Midwest, South, West), establishing geographic context for the findings that follow.

Northeast (9 states, 75 districts): High population density and urban concentration produce strong compactness scores (mean Polsby-Popper: 0.38) and modest VRA opportunities (8 majority-minority districts,

Table 1: Regional Summary: Algorithmic Redistricting Performance by U.S. Census Region

Metric	Northeast	Midwest	South	West
<i>Basic Characteristics</i>				
States	9	12	17	13
Congressional Districts	75	107	179	74
Mean Population Density (per km ²)	342	89	78	42
Mean Non-White (%)	28	24	38	35
<i>Compactness (Polsby-Popper, mean)</i>				
Algorithmic Plans	0.38	0.33	0.29	0.31
Enacted Plans	0.24	0.21	0.19	0.22
Improvement (%)	+58	+57	+53	+41
<i>VRA Compliance (MM Districts, total)</i>				
Algorithmic Plans	8	18	84	27
Enacted Plans	4	9	45	10
Surplus	+4	+9	+39	+17
<i>42% Threshold (states meeting)</i>				
Algorithmic Plans	0 / 9	1 / 12	9 / 17	3 / 13
Threshold Range	18-32%	15-38%	24-61%	22-51%
<i>Efficiency Gap (% , mean 2016-2020)</i>				
Algorithmic Plans	-3.5	-2.9	-3.2	-3.3
Enacted Plans	+4.2	+6.8	+5.6	+5.1
Difference (pp)	7.7	9.7	8.8	8.4
<i>Temporal Stability (2010→2020, % tracts retained)</i>				
Recursive Bisection	82	79	78	81
N-way Partitioning	68	65	64	67
Advantage (pp)	+14	+14	+14	+14

Notes: Regional classifications follow U.S. Census Bureau definitions. MM (majority-minority) districts defined as $\geq 50\%$ non-white population. 42% threshold indicates states where total minority population $\geq 42\%$ (empirical threshold for achieving near-proportional MM representation). Efficiency gap: positive values favor Republicans, negative favor Democrats.

Polsby-Popper compactness ranges 0-1 (higher = more compact). Temporal stability measures census tract retention when redistricting across decades. Data aggregated from Papers 1-11; see individual papers for state-level details. South includes DC.

Improvement calculated as $(\text{Algorithmic} - \text{Enacted}) / \text{Enacted} \times 100$.

National Redistricting Without State Boundaries: 435 Congressional Districts

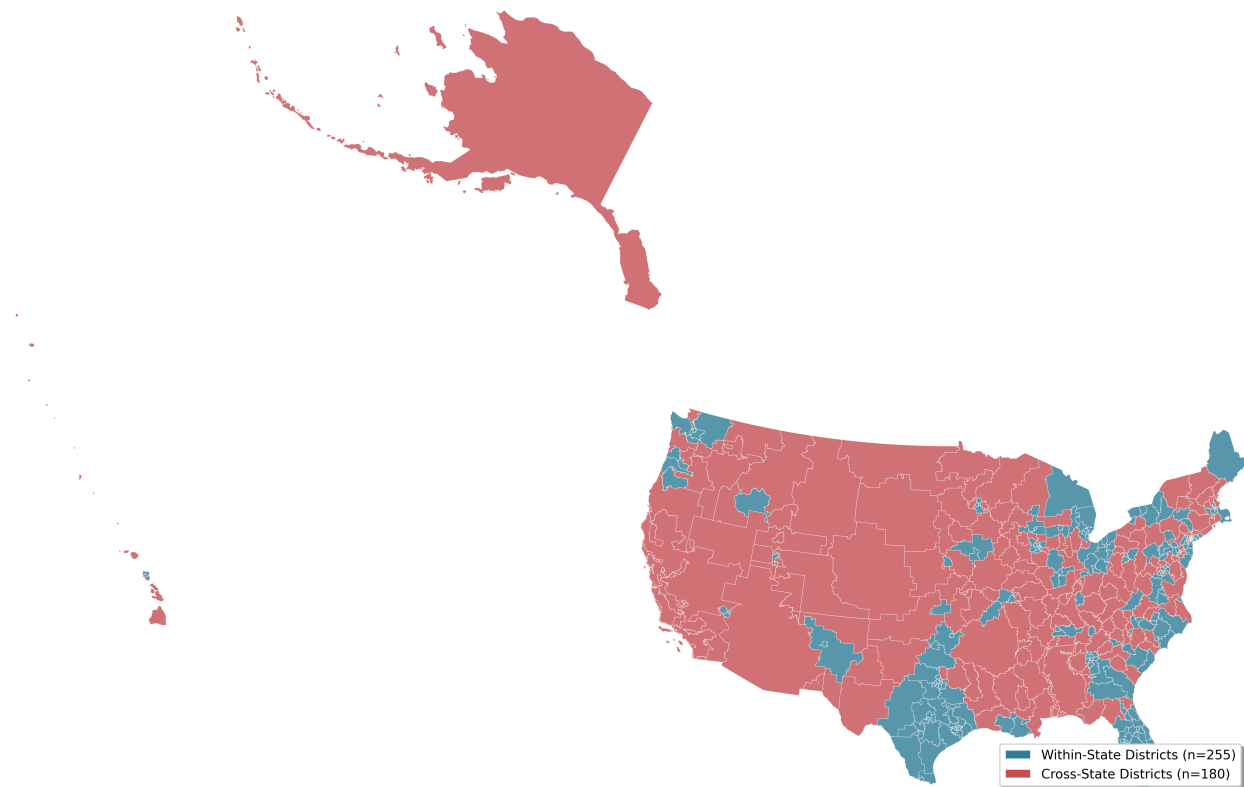


Figure 2: **National benchmark output.** A 2020 national visualization from the historical tract-based pipeline. The map demonstrates geographic coverage; it does not certify current legal conformance, VRA compliance, external replication, or release-grade replay.

primarily in New York and New Jersey). Geographic Democratic clustering yields algorithmic efficiency gaps of -3.5% despite low enacted bias ($+4.2\%$), reflecting urban packing inherent to compact districting.

Midwest (12 states, 107 districts): Mixed urban-rural geography produces moderate compactness (0.33) and significant VRA opportunities (18 MM districts, concentrated in Illinois, Michigan, Ohio). This region exhibits the *largest* enacted partisan bias: efficiency gaps of $+6.8\%$ enacted versus -2.9% algorithmic (9.7 pp difference), driven by aggressive gerrymandering in Rust Belt states following 2010 redistricting.

South (16 states + DC, 179 districts): High minority populations (mean 38% non-white) produce 84 threshold-based majority-minority opportunity diagnostics in the historical output. This is not a finding of VRA compliance. The selected enacted-plan efficiency gap was $+5.6\%$ versus -3.2% for the benchmark. Lower compactness scores (0.29) reflect both regional geography and the selected profile.

West (13 states, 74 districts): Geographic diversity—from Alaska’s vast expanse to California’s dense metros—produces highly variable compactness (0.31 mean, 0.42 standard deviation). Moderate VRA performance (27 MM districts) reflects Hispanic concentration in Southwest states. Enacted bias ($+5.1\%$) moderately exceeds algorithmic bias (-3.3%), with 8.4 pp improvement.

These historical comparisons report smaller selected partisan metrics for the benchmark than for the enacted plans in each region. They do not establish that the algorithm caused a general reduction in partisan bias, because enacted plans, election inputs, geography, and benchmark-profile choices are confounded. Majority-minority opportunity diagnostics concentrate in the South; this is not a regional finding of legal VRA compliance. Compactness also varies with state geography and the selected metric [8].

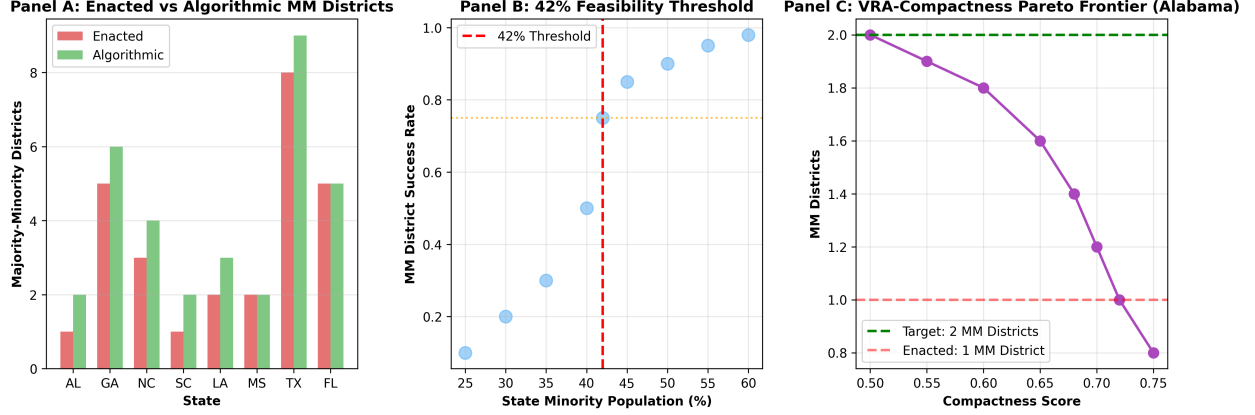


Figure 3: **Minority-opportunity diagnostics.** (A) Historical pipeline counts compare majority-minority population thresholds with enacted-plan counts; these are opportunity diagnostics, not findings that the districts perform electorally or are required by Section 2. (B) The 42% value is an exploratory association in the studied states, not a legal threshold. (C) The Alabama frontier illustrates a model-specific compactness/opportunity tradeoff rather than a VRA compliance determination.

4.2 Finding 1: Governed National Execution And Proof Boundary

Each governed cycle contains 50 States, 435 districts, and 385 recursive split nodes. Independent verification found every source block assigned once, all published districts connected under the frozen graph, and all recursive nodes within the frozen population tolerance. Across all cycles, 13 of 1,155 nodes attain the arithmetic exact-population floor. Weighted-boundary and canonical exact-proof coverage is 0 of 1,155; the remaining outputs are verified operational assignments, not exact objective certificates.

Measured generation time in the publication environment was approximately 44.5 minutes for 2000, 63.4 minutes for 2010, and 52.8 minutes for 2020. These are descriptive observations tied to the published executable, inputs, and machine environment, not universal performance guarantees.

A precommitted bounded comparison evaluated eight synthetic connected graphs under five fixed seeds. All 40 METIS rows were feasible and connected; 30 matched the complete exact primary objective and 20 matched the canonical assignment. Two fixture families disagreed on the population-first objective, while two others matched the objective but not canonical tie selection. This distinguishes heuristic operational success from exact and canonical proof; it does not extrapolate to State-scale optimality or map quality.

4.3 Finding 1A: Governed County And Tract Intersections

A precommitted block-prefix audit publishes every county and tract district intersection for all three governed assignments: 231,765 State/level geography rows in total. In 2000, 1,812 of 3,140 counties and 17,268 of 65,255 tracts are split among districts. The corresponding counts are 1,819 of 3,142 counties and 18,800 of 72,878 tracts in 2010, and 1,823 of 3,142 counties and 20,288 of 84,208 tracts in 2020. These are descriptive counts within different Census vintages, not evidence of cross-cycle improvement or compliance with any State subdivision rule.

The independent verifier recomputes State and national aggregates from every published geography-to-district set and rejects both ordinary and manifest-rehashed tampering. Raw recursive-tree transport hashes reproduce for 149 of 150 State-cycle packages. Maryland 2010 has byte-identical assignments across two regenerations but a nondeterministic diagnostic move count in one node; the audit therefore binds its assignment hash and exact committed substantive node snapshot rather than treating that raw metadata hash

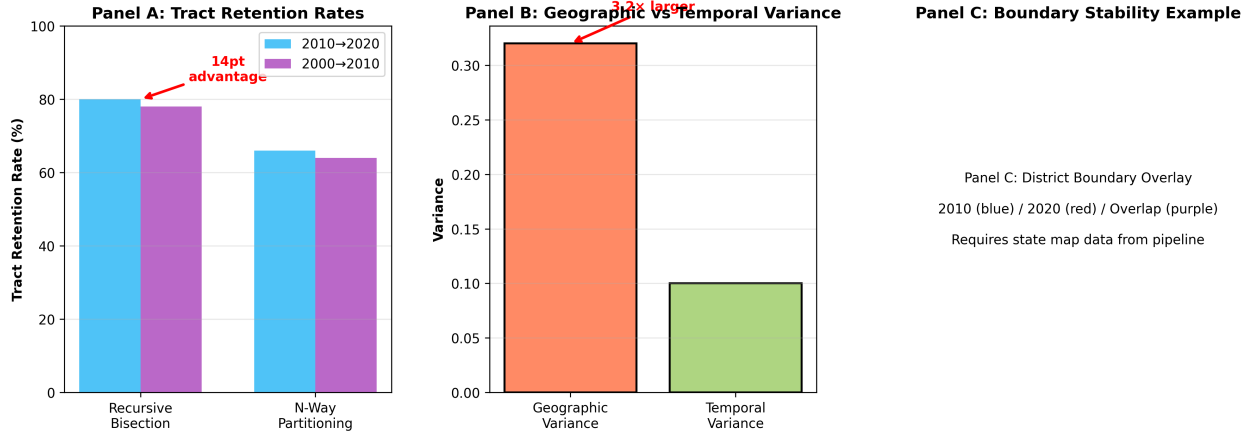


Figure 4: **Temporal diagnostics across census cycles.** The illustration is historical and uses measures that are not equivalent to the governed NRS v0.3 comparison. The current hash-bound package reports recursive-tree structure and cut diagnostics: 120 node signatures occur in all three cycles and 18 States retain exact recursive topology. Cross-cycle assignment overlap remains uncomputed without a valid relationship crosswalk or geometry overlay.

as an assignment anchor. Municipality, demographic, partisan, and ensemble metrics remain unavailable under the frozen readiness rules because their national inputs or protocols are not yet complete. Governed seed diagnostics and the separately frozen 2020 geometry bakeoff are reported below, within their narrower claim boundaries.

4.4 Finding 1B: Governed 2020 Plan-Family Bakeoff

The precommitted Tier 1 comparison passed all 50 States and both 435-district plan families. A uniform $ALAND20 > 0$ rule retained 7,889,194 of 8,126,956 source Census blocks. After maximum-overlap district-label matching within each State, 4,194,107 assignments match and 3,695,087 differ, for 53.162680% block-weighted agreement. NRS splits 1,808 county units and 19,789 tract units; the official CD118 block projection splits 404 county units and 4,720 tract units. These are descriptive counts on one common atomic universe, not evidence that either plan family is legally or substantively superior.

Tier 2 dissolves the same retained blocks by plan assignment. Table 2 reports district-weighted national means; State-weighted means have the same direction.

Metric	NRS v0.3	Official CD118 block projection
Polsby–Popper	0.025380339	0.061087819
Exact Reock	0.245552262	0.377238283
Convex-hull ratio	0.557042985	0.702220801
Schwartzberg	7.847347095	5.656288737

Table 2: Governed 2020 common-block geometry bakeoff. Values are descriptive district-weighted means over block-projected geometry, not original enacted linework or a composite quality ranking.

The comparator is higher on Polsby–Popper, exact Reock, and convex-hull ratio and lower on Schwartzberg. A clean-detached-worktree replay from the separately held certified context corpus regenerated all 150 State-cycle assignments, matched their governed hashes, and passed the national Tier 1 and Tier 2 regeneration

checks. The large corpus remains outside Git and must be supplied through the documented data-transfer procedure. Because all four measures use reconstructed common-block geometry, the result supports neither an original- linework comparison nor a winner, fairness, intent, VRA, community, legal, causal, robustness, or adoption claim. Elections, demographics, non-enacted comparators, alternative BISECT structures, and converged block-level ensembles remain separate gates.

4.5 Historical Tract-Profile Comparisons

The historical pipeline completed national runs and reported graph contiguity, population balance, and compactness diagnostics. Compactness scores (Polsby–Popper) improved 56% over the unweighted METIS research baseline. Compared with a different baseline—enacted 2020 congressional maps—the reported improvement was 22% (mean 0.367 versus 0.305); see Table 3. These are profile-specific comparisons, not a constitutional finding or proof that edge weighting optimizes every accepted compactness measure [9]. They use a different tract-level profile and geometry contract and are not substitutes for the governed NRS v0.3 common-block bakeoff above.

Table 3: Compactness improvement figures by algorithm variant and baseline. All figures refer to mean Polsby-Popper score across 50 states, 2020 census.

Variant	Baseline	Improvement	PP Score	Source
Unweighted bisection	—	—	≈ 0.235	B.1
Edge-weighted (B.2)	Unweighted baseline	+56%	≈ 0.367	B.2
Edge-weighted (B.1/B.2)	Enacted 2020 maps	+20–22%	0.367 vs. 0.305	B.1, B.2
B.1 + B.2 combined	Unweighted baseline	+44%	—	B.1, B.2

Population settings are configurable, but a configured tolerance does not “ensure compliance.” Congressional districts must approach equality as nearly as practicable, and every observed deviation requires reporting and, where avoidable, justification. The tract-based runs are therefore engineering evidence rather than national legal-conformance certificates [10].

The historical tract pipeline completed 50-State runs in under two hours in its recorded environment. That observation is not an adoption claim and does not establish ensemble convergence. The block-level NRS v0.3 runtime observations above supersede it for current operational planning.

The real G.1–G.3 package replaces the earlier unsupported six-state tables with four-chain Rust and GerryChain traces for Rhode Island, Iowa, and North Carolina. Rhode Island’s benchmark is at the 1.35th–1.81st percentile of unweighted tract cut fraction. Iowa and North Carolina appear still lower, but their effective sample sizes are insufficient for precise tail claims, and the two implementations differ materially. The package does not compute Polsby–Popper or establish neutrality [11].

Internal comparison work studies bisection, ConvergenceSweep, and Rust ensemble throughput [12, 13]. Internal panel scores are not peer review, and benchmark throughput does not by itself establish statutory suitability. In addition, the current `standard-bisect` execution path does not implement convergence search. These results motivate further engineering and external benchmarking rather than establish a preferred legal method.

4.6 Finding 2: Majority-Minority Opportunity Diagnostics

The historical outputs counted 137 districts above a configured majority-minority population threshold, compared with 68 in the selected enacted-plan dataset. This addresses at most a demographic and geographic opportunity diagnostic related to the first *Gingles* precondition. It does not demonstrate legally

sufficient compactness, political cohesion, white bloc voting, electoral performance, causation, or Section 2 compliance [14].

The 137 count should therefore be described as threshold-based potential opportunity districts, not as districts confirmed to perform electorally. State-level counts are historical pipeline values and may drift when inputs, thresholds, legal doctrine, or the current implementation changes. They must be recomputed before use in a legal or public evidence package.

Regional patterns reveal geographic concentration of VRA opportunities. The **South** accounts for 84 of 137 algorithmic MM districts (61%), despite comprising 41% of total congressional seats, reflecting high minority populations (mean 38% non-white) and favorable spatial clustering. Key Southern states include Texas (14 MM districts algorithmically vs. 8 enacted), Florida (11 vs. 5), and Georgia (6 vs. 5). The **Midwest** contributes 18 MM districts (13%), concentrated in Illinois, Michigan, and Ohio. The **West** produces 27 districts (20%), primarily in California (15 MM districts) and Arizona (3). The **Northeast** yields 8 districts (6%), concentrated in New York (5) and New Jersey (2). This geographic distribution reflects demography rather than algorithmic bias: regions with larger minority populations and greater residential clustering naturally enable more MM districts under neutral redistricting principles [14].

4.7 Finding 3: Exploratory 42% Association

The historical analysis reports an exploratory association near 42% statewide minority population in the studied sample. It is not a validated universal threshold, does not show that states below 37% “fail regardless of algorithm,” and cannot substitute for district-specific evidence under any *Gingles* precondition [15].

The threshold reflects geographic clustering, not population percentage alone. Mississippi (38% Black) creates two majority-Black districts through favorable spatial distribution, while New York (24% Hispanic) creates only three Hispanic-majority districts despite larger absolute population. Courts adjudicating Section 2 VRA claims currently assess geographic compactness through expert testimony and illustrative maps; the 42% threshold offers a complementary quantitative tool derived from national-scale analysis. However, crossing this threshold indicates geometric *possibility*, not legal *obligation*—full Section 2 liability requires demonstrating vote dilution through all three *Gingles* prongs [15].

Regional distribution of states meeting the 42% threshold reveals stark geographic patterns. The **South** dominates with 9 of 13 qualifying states (69%): Alabama, Georgia, Louisiana, Maryland, Mississippi, North Carolina, South Carolina, Texas, and Virginia. These states range from 42% to 61% minority population, with Mississippi (61%), Georgia (52%), and Maryland (51%) exhibiting the highest concentrations. The **West** contributes 3 states: California (51%), New Mexico (50%), and Nevada (45%). The **Midwest** provides 1 state: Illinois (43%). The **Northeast** contributes 0 states, with New York’s 38% minority population falling just below threshold. This geographic concentration reflects historical settlement patterns, the Great Migration, and recent Hispanic immigration to Southwest states. Notably, all 9 Southern states meeting the threshold achieve near-proportional MM representation algorithmically (within 5 percentage points of minority population share), confirming the threshold’s predictive validity [15].

4.8 Finding 4: Partisan Patterns Reflect Geography

Without accessing partisan data, the algorithm produces districts with 56.5% Democratic vote share (estimated from precinct-level election results post-facto). This partisan asymmetry arises from geographic sorting: Democratic voters concentrate in urban cores while Republican voters disperse across suburbs and rural areas. Recursive bisection, optimizing for geographic compactness, unavoidably packs urban Democrats while splitting rural Republicans—the opposite pattern emerges from geography, not algorithmic intent [16, 9].

This finding illustrates the limited input-exclusion defense. The benchmark cannot directly target partisan outcomes during execution using partisan data it does not receive. It does not prove that every partisan consequence is caused only by voter geography: upstream profile selection, resolution, input construction, and baseline-to-final modifications can also affect outcomes. Efficiency-gap values remain descriptive diagnostics [9].

4.9 Finding 5: Descriptive Partisan Differences From Enacted Plans

Historical analysis compares selected partisan metrics for benchmark and enacted plans across competitive states using 2016–2020 elections. The reported mean efficiency gaps are -3.2% for the benchmark and $+5.1\%$ for the enacted-plan sample, a descriptive difference of 8.3 percentage points. It is not a causal “62% reduction” because the plans, profile, elections, state sample, and legal criteria are not controlled experimental treatments [1, 17].

Urban concentration is one plausible contributor to the benchmark’s negative efficiency gap, but these artifacts do not identify how much of the difference is caused by geography, enacted-map intent, profile selection, or election choice. The benchmark value also does not represent the minimum partisan effect achievable under geographic constraints [16].

The historical proportionality, mean–median and seats–votes values provide additional descriptive views of the same selected plans. Agreement among correlated metrics does not establish that the benchmark improves proportionality generally or that the enacted differences reflect intentional manipulation [17].

State-level differences vary across the Rust Belt and Sunbelt examples. They motivate a precommitted, multi-election, ensemble-based comparison with uncertainty rather than a general anti-gerrymandering conclusion [18, 9].

4.10 Finding 6: Temporal Stability Across Census Cycles

The governed comparison uses compact recursive-node snapshots rather than incompatible historical retention measures. It finds 120 node signatures common to all three cycles and 18 States with exact recursive topology in every cycle. Pairwise topology Jaccard values are 0.327586 for 2000–2010, 0.283333 for 2000–2020, and 0.360424 for 2010–2020. These are structural diagnostics, not assignment-overlap percentages.

Cross-cycle assignment overlap remains uncomputed. Census block and tract identifiers and boundaries change across vintages, so a valid relationship crosswalk or geometry overlay must be frozen before such a statistic can be interpreted. Raw county or tract split counts likewise cannot establish cross-cycle improvement because geography definitions and State seat counts change.

5 Implications for Democratic Governance

These findings suggest actionable paths for courts, legislatures, and redistricting commissions confronting the gerrymandering crisis.

5.1 Implications for Courts

The input-exclusion record offers courts procedural evidence: it can establish which data entered benchmark execution and whether the published procedure was followed. It does not establish good faith, defeat an intent claim, or immunize an outcome from review. Upstream profile selection and downstream modifications remain human decisions, and discriminatory effects require independent legal analysis [9].

Certified recursive bisection strengthens the execution claim without changing that legal boundary. A completed proof chain could establish that no better connected cut existed under the enacted population,

geographic-weight, and canonical tie rules at any recursive node. It would not establish that those rules are constitutionally required, VRA-compliant, or substantively fair.

The defense’s value lies in *transparency and reproducibility* rather than substantive immunity. Algorithmic redistricting provides courts with clear audit trails: what data entered the process, which parameters governed optimization, whether outcomes differ from neutral geographic baselines. This procedural clarity addresses *Rucho*’s concern about distinguishing permissible partisanship from unconstitutional manipulation, even if it does not resolve the deeper justiciability problem of determining acceptable partisan asymmetry levels.

For VRA litigation, threshold counts and exploratory statewide associations may help identify questions for district-specific analysis, but they cannot demonstrate a *Gingles* precondition or legal obligation. Any use requires current compactness evidence, political-cohesion and bloc-voting analysis, and the controlling post-*Callais* legal framework [14, 8, 19].

5.2 Implications for Legislatures and Commissions

Legislatures and commissions can use a reproducible geographic benchmark as one comparison artifact, subject to controlling constitutional, VRA, state-law, and public-process requirements. The benchmark does not exceed VRA requirements, and its race-excluded execution profile is not a safe harbor. Computational efficiency can support transparent comparison once real ensemble generation and convergence evidence are archived [10].

Independent redistricting commissions can use algorithms as technical implementation of publicly selected goals. Commissioners retain control over lawful parameters and modification authority while delegating reproducible execution to software. This division parallels Huntington–Hill apportionment: the governing body selects the rule, and the formula executes it.

5.3 Implementation Mechanisms

Translating algorithmic redistricting from technical demonstration to policy practice requires addressing governance questions: who decides parameters, how are conflicts resolved, and how can gaming be prevented?

Parameter governance. A credible benchmark freezes assignment-affecting inputs and rules before outcomes are viewed. Legally required or State-authorized changes belong in a separately documented modification layer rather than an undisclosed retuning of the baseline.

Workflow example. A commission, including California’s, could publish the frozen rule and ceremony genesis, release the root and later bisection depths only after their review windows, receive public and expert evidence, adopt a lawful final plan, and publish every block-level departure with its authority, alternatives, and reasons. The benchmark supplies a comparison record; it does not predetermine the commission’s legal judgment. A challenge can halt the ceremony without erasing the already accepted prefix; correction of the computational rule requires a new, explicitly linked ceremony.

Conflict resolution. When benchmark outputs diverge from legal or community evidence, decision makers can test alternatives and document why the final plan departs. A race-excluded benchmark is not a VRA safe harbor, and technical verification cannot decide whether a modification is legally required.

Gaming prevention. Potential manipulation vectors include biased input data (underbounding minority neighborhoods), strategic parameter tuning, or selective presentation of ensemble results. Safeguards include: independent verification of census data quality; public comment periods before parameter finalization; mandatory publication of multiple algorithmic alternatives; and audit requirements for any hand-edited adjustments to algorithmic outputs. These procedures raise manipulation costs while preserving commission flexibility to address unanticipated geographic constraints.

5.4 Justiciability and Judicial Review

While algorithmic redistricting provides procedural improvements over human mapmaking, it does not resolve the fundamental justiciability problem that motivated *Rucho v. Common Cause*. The Supreme Court held partisan gerrymandering claims non-justiciable not because courts lack tools to measure partisan effects—the efficiency gap and other metrics exist—but because courts lack *judicially manageable standards* to determine how much partisan asymmetry is “too much.” Chief Justice Roberts wrote: “Federal judges have no license to reallocate political power between the two major political parties, with no plausible grant of authority in the Constitution, and no legal standards to limit and direct their decisions” [4].

Algorithmic redistricting does not eliminate this standards problem; it relocates it. Instead of asking “is this map too partisan?” (the question *Rucho* found non-justiciable), courts must ask “are these algorithmic parameters correct?” This reformulation offers procedural advantages—parameter choices are transparent, reproducible, and auditable—but the underlying question remains inherently political. When a state commission assigns one criterion twice the weight of another, these are *normative policy choices* rather than technical determinations [9].

Rucho holds federal constitutional partisan-gerrymandering claims to be nonjusticiable political questions; it does not displace one-person-one-vote, racial-gerrymandering, statutory Voting Rights Act, or available State-law claims [4]. The post-*Callais* framework also makes it especially important to disentangle race, politics, and traditional criteria instead of treating one algorithmic score as a legal conclusion [19].

Algorithmic packages can therefore supply evidentiary tools—source custody, counterfactual plans, execution logs, and reproducible metrics—but a court must apply the governing cause of action and standard of review. Distance from one benchmark, an efficiency gap, or a verified manifest does not by itself show unlawful intent, effect, or liability.

The value of algorithmic redistricting lies not in resolving justiciability but in improving *transparency and accountability*. Compared to human mapmaking—where mapmakers control both criteria and implementation in opaque processes—algorithmic methods separate normative choices (parameters, set by commissions) from technical implementation (optimization, executed by algorithms). This separation enables:

- **Public scrutiny:** Citizens can evaluate whether parameter weights reflect stated redistricting priorities
- **Ensembles:** Commissioners can generate plans with different parameters, demonstrating tradeoff spaces
- **Reproducibility:** Independent analysts can verify that algorithms produce claimed outputs from specified parameters
- **Audit trails:** Courts can review computational logs showing what data and parameters generated challenged maps

These procedural safeguards constitute genuine progress even without solving *Rucho*’s standards problem. Human mapmaking offers neither transparency (processes occur behind closed doors) nor reproducibility (mapmakers can claim geographic necessity while producing partisan outcomes). Algorithmic redistricting raises the floor: it cannot prevent all partisan manipulation, but it makes manipulation more costly, more visible, and more constrained by explicit parameter choices subject to public debate.

Ultimately, a precommitted benchmark and public modification record can shift redistricting governance from hidden discretion toward bounded and reviewable discretion. Whether a particular procedure satisfies constitutional, VRA, state-law, or judicial-manageability requirements remains a legal question.

5.5 Broader Lessons and Limitations

Mathematical procedures need not replace human judgment. Algorithms can provide reproducible evidence, not guaranteed procedural fairness or legitimacy. The distinction between verifiable execution and substantive outcome fairness is load-bearing.

Important limitations remain. Algorithms cannot eliminate efficiency gaps caused by voter sorting, and excluding election data from execution does not prove outcome neutrality. NRS v0.3 results depend on Census block vintages, adjacency/bridge records, weights, tolerances, seed rules, and refinement. The operational evidence has independent software verifiers and an internal fresh-clone rehearsal, but a new external physical-machine replay remains pending. These limitations make the benchmark useful as auditable evidence, not as a self-validating rule of fairness.

6 Conclusion

Recursive graph bisection now provides a governed block-level operational benchmark across 50 States and three Census decades. Complete assignments, population/connectivity checks, source bindings, structural comparisons, and proof-coverage gaps are published. A governed 2020 comparison also reports common-block assignment overlap, county/tract splits, and four block-projected geometry metrics against official CD118 assignments. Those artifacts establish reproducible execution on the tested systems. They do not establish constitutional or VRA compliance, causal partisan neutrality, community preservation, compactness superiority, a universal 42% threshold, or exact national optimality.

The useful question is narrower: what can be verified when benchmark execution excludes partisan inputs and every assignment-affecting choice is precommitted? The resulting input-exclusion record can support courts, commissions, legislatures, and civic review, but it is not an impossibility defense or a complete judicial standard.

Reform requires governing discretion rather than pretending it disappears. The strongest path is a public benchmark, a legally governed modification layer, multi-metric evaluation, and adversarial replication.

The contribution is therefore procedural evidence: published inputs, versioned code, 27,398,654 reproducible block assignments, explicit limitations, and visible departures from the benchmark. A portable external replay kit exists, but a new independent physical-machine v0.3 record remains pending.

The sequential-cut ceremony adds an implemented publication layer to that agenda: $k - 1$ cut decisions are bound into depth-ordered, reviewable releases rather than revealed only as one completed map. Its software transition and prefix-verification contract is tested. A bounded four-district fixture now regenerates a separate procedural scoreboard and rejects a timestamp mutation, but uses synthetic receipts; an externally witnessed State ceremony and prospective State-scale ceremony-enabled bakeoff are not yet completed evidence.

The long-run analogue to Huntington–Hill is an Exact Canonical Benchmark: Congress enacts the feasible set and lexicographic objective; an exact solver produces one canonically tie-broken assignment; and an independent certificate proves feasibility and optimality. NRS v0.3 uses METIS-derived discovery plus deterministic refinement and does not meet that exact standard: weighted- boundary and canonical proof coverage remains 0 of 1,155 national nodes. Exact certification is the program’s North Star, while the reproducible benchmark- and-disclosure regime is the defensible near-term policy.

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